Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC II Business Statistics Examination May 2017

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate Part II (Class XII) Business Statistics. It contains comments on candidates' responses to the 2017 HSSC II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on students' performance on every question and *some* specific examples of students' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

Key observations:

This year candidates did not perform well in questions based on price index, permutation and combination were not well attempted. Weaker responses revealed that candidates had problems with finding median and mode for grouped data and probability.

Constructed Response Questions (CRQs)

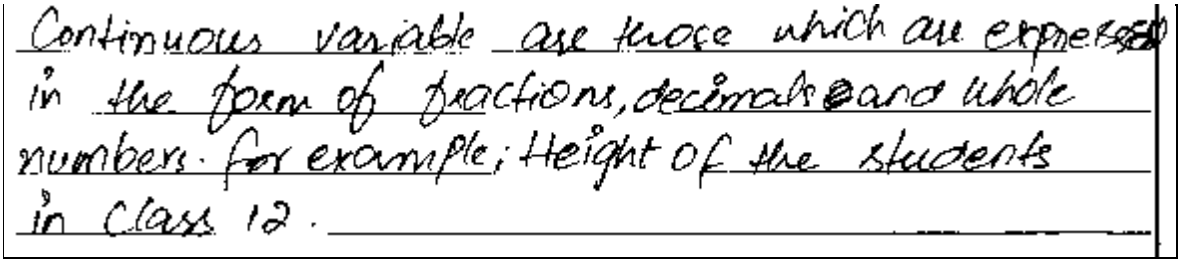
Detailed Comments:

Question 1:
It was a well attempted question.

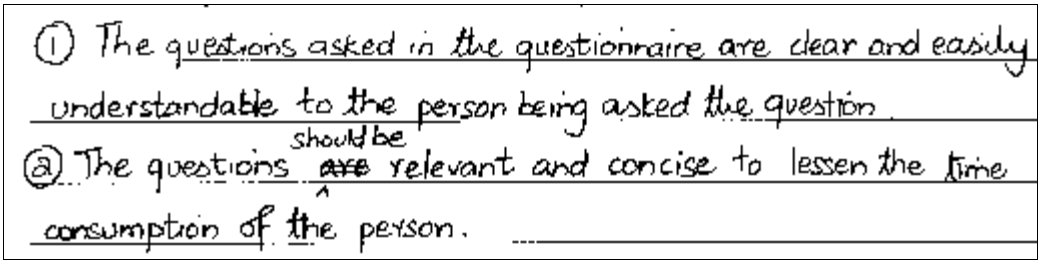
Question 1:
i. Define continuous variable and give ONE example of it.
ii. Write any TWO characteristics of a good questionnaire.

Better responses indicated that in part i, the candidates were able to express what a continuous variable is. In part ii, candidates provided a wide range of characteristics of good questionnaire.

Example: i.

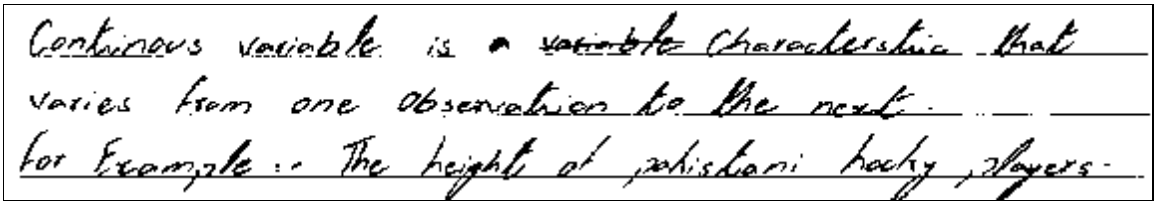

Continuous variable are those which are expressed in the form of fractions, decimals and whole numbers. for example; Height of the students in class 12.

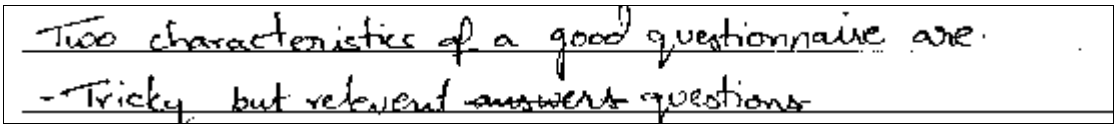
Example: ii.


① The questions asked in the questionnaire are clear and easily understandable to the person being asked the question.
② The questions ~~are~~ should be relevant and concise to lessen the time consumption of the person.

Weaker responses revealed that in part i, candidates lacked knowledge of continuous variable and confused it with discreet variable. Part ii reflected that candidates had no idea what a questionnaire is and wrote random details.

Example: i


Continous variable is a ~~variable~~ Characterstic that varies from one observation to the next. for Example :- The height of pakistani hocky players.


Two characteristics of a good questionnaire are.
- Tricky but relevent ~~answers~~ questions

Question 2:

It was a well attempted question.

Question 2:

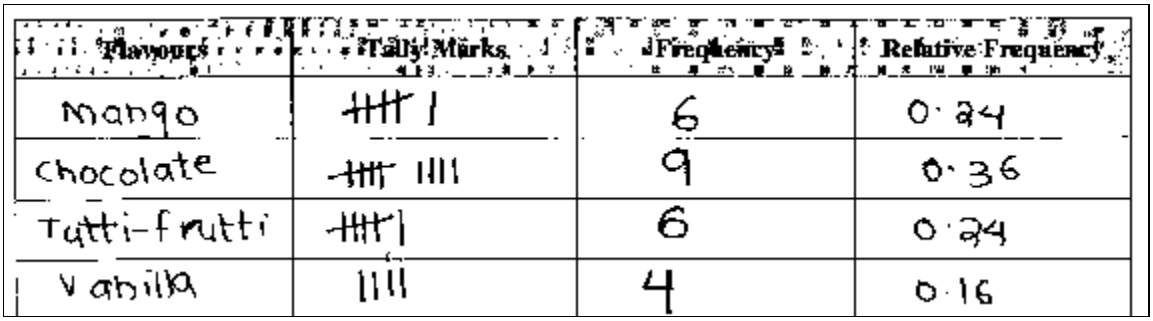
Mr Amjad asked the students of his class about their favourite ice cream flavour. The students' responses are given in the following grid.

Mango	Vanilla	Mango	Chocolate	Chocolate
Vanilla	Mango	Chocolate	Chocolate	Chocolate
Tutti-frutti	Vanilla	Chocolate	Mango	Tutti-frutti
Chocolate	Tutti-frutti	Chocolate	Tutti-frutti	Tutti-frutti
Tutti-frutti	Vanilla	Mango	Chocolate	Mango

Fill the given table to construct the frequency distribution of the data given in the grid. Also find the relative frequency for each flavour.

Better responses revealed that the candidates understood how to convert ungrouped data to grouped data. The tally marks were used to find the frequency. Relative frequency was found in the end in terms of ratio and percentage both.

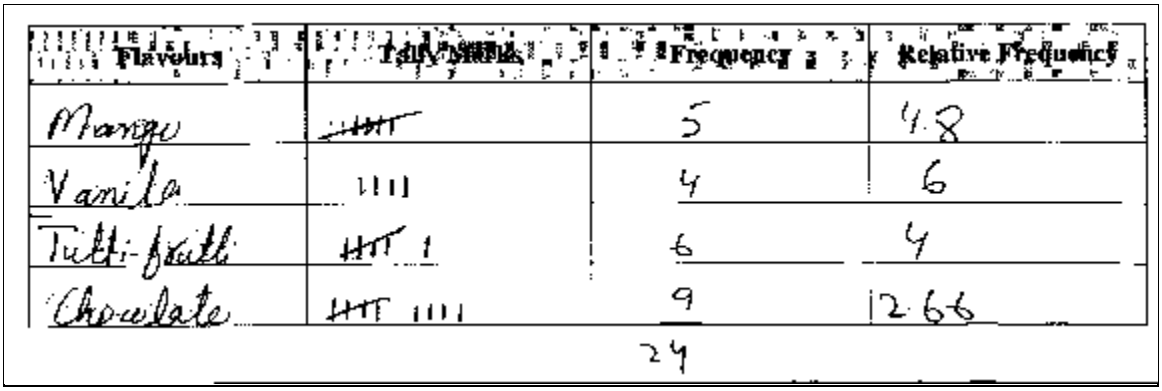
Example:



Flavours	Tally Marks	Frequency	Relative Frequency
Mango	𝍸 I	6	0·24
Chocolate	𝍸 IIII	9	0·36
Tutti-frutti	𝍸 I	6	0·24
Vanilla	IIII	4	0·16

Weaker responses displayed that candidates made minor errors in making tally marks due to which they got incorrect frequency. Many candidates did not know what relative frequency is and found random ratios.

Example:



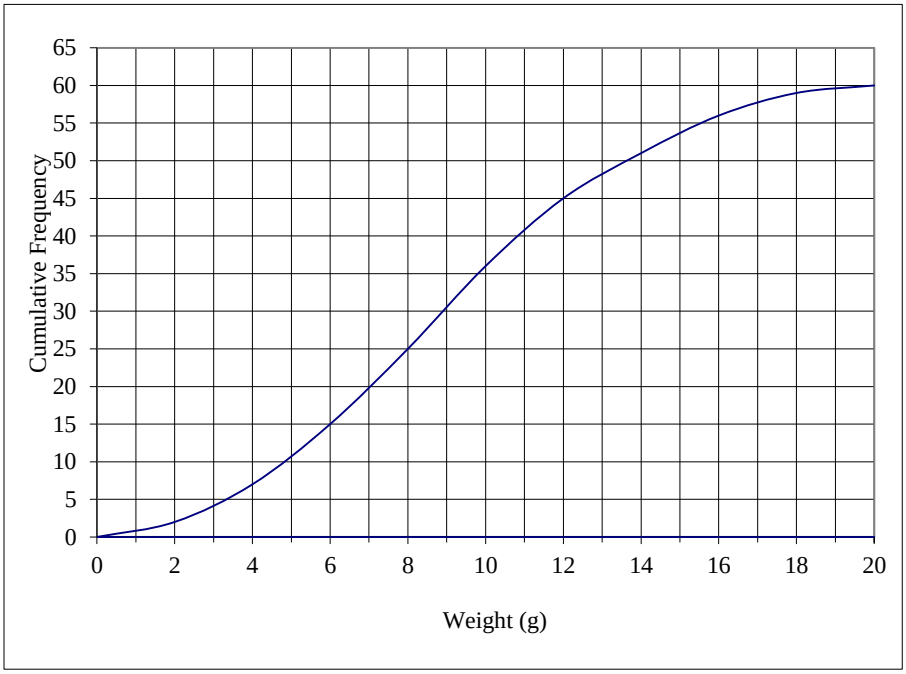
Flavours	Tally Marks	Frequency	Relative Frequency
Mango	𝍸	5	4.8
Vanila	IIII	4	6
Tutti-fruti	𝍸 I	6	4
Chocolate	𝍸 IIII	9	12.66
		24	

Question 3:

This question showed average performance.

Question 3:

The given cumulative frequency curve shows the weight (in grams) of 60 packages to be delivered by a courier company.



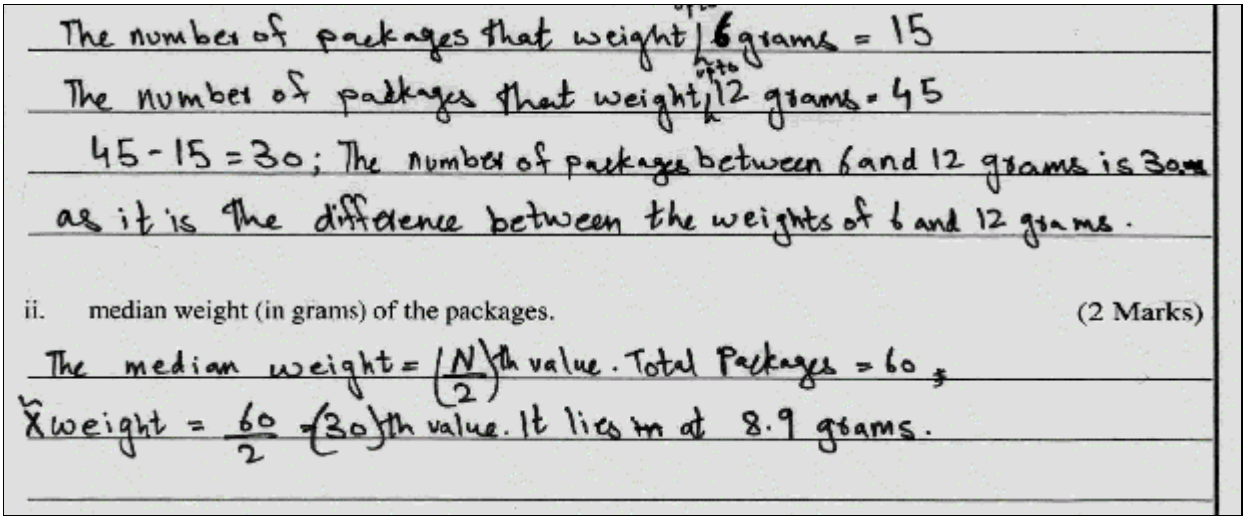
Using the given cumulative frequency curve, estimate the

i. number of packages that weigh between 6 grams to 12 grams.

ii. median weight (in grams) of the packages.

Better responses reflected that candidates were able to interpret cumulative frequency curve. For number of packages between 6 to 12 grams, candidates first found the number of packages less than or equal to 12 grams and then subtract it from the number of packages less than 6 grams.

Example:


The number of packages that weight 6 grams = 15

The number of packages that weight 12 grams = 45

45 - 15 = 30; The number of packages between 6 and 12 grams is 30 as it is the difference between the weights of 6 and 12 grams.

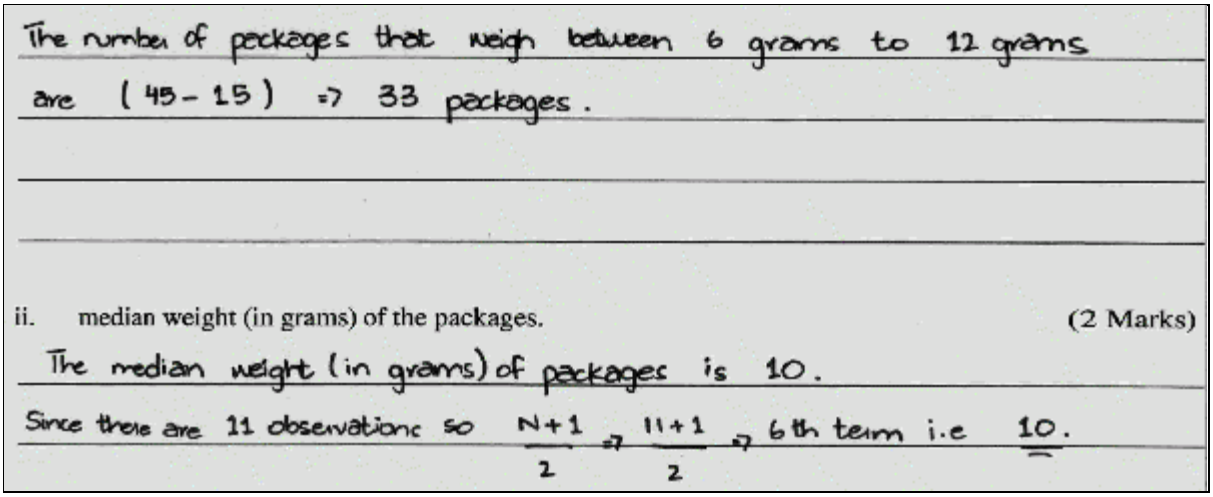
ii. median weight (in grams) of the packages. (2 Marks)

The median weight = $\left(\frac{N}{2}\right)$th value. Total Packages = 60

$\tilde{X}$ weight = $\frac{60}{2}$ = (30)th value. It lies in at 8.9 grams.

Weaker responses indicated that candidates had difficulty interpreting the cumulative frequency graph. Instead of using graph to answer the question as instructed, candidates used different formulae. Many candidates made mistakes in reading the graph. For finding median weight, candidates found 50% of the frequency and took that as the answer.

Example:


The number of peckages that weigh between 6 grams to 12 grams are (45 - 15) => 33 packages.

ii. median weight (in grams) of the packages. (2 Marks)

The median weight (in grams) of packages is 10.

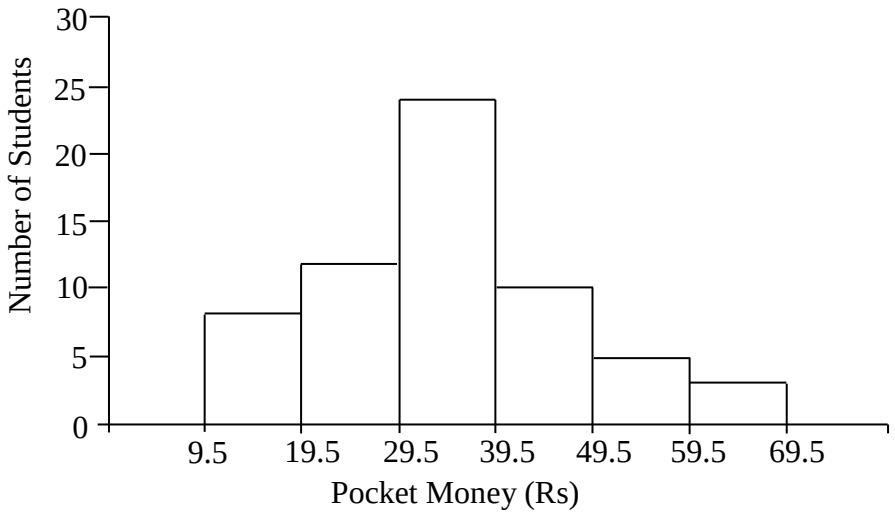
Since there are 11 observations so $\frac{N+1}{2}$ => $\frac{11+1}{2}$ => 6th term i.e 10.

Question 4:

This question offered a choice between part a and part b. Majority of the candidates attempted part a. This was generally a well attempted question.

Question 4a:

The diagram shows a histogram which represents the daily pocket money (in Rupees) of students in a class.



i. Estimate the mode of this data graphically.

ii. The frequency distribution of this histogram is shown in the given table. Fill in the table appropriately to find the mean of the data.

Class Boundary	**Number of Students**	**Class Mark**	
9.5 – 19.5	9	14.5	
19.5 – 29.5	12	24.5	
29.5 – 39.5	24	34.5	
39.5 – 49.5	10	44.5	
49.5 – 59.5	5	54.5	
59.5 – 69.5	3	64.5	
Total	63	—	

iii. Using the mode and mean of the data found in part (i) and (ii), estimate the median of the data. (**Hint:** Use empirical relationship)

Better responses reflected that candidates were able to find the mode graphically from histogram. The table was completed using the given histogram and the mean was found. In part ii, candidates used the empirical formula to find the answer.

Example:

i. Estimate the mode of this data graphically. (1 Mark)

Approx. 34.5

ii. The frequency distribution of this histogram is shown in the given table. Fill in the table appropriately to find the mean of the data. (2 Marks)

Class Boundary	Number of Students	Class Mark	fx
9.5 – 19.5	9	14.5	130.5
19.5 – 29.5	12	24.5	294
29.5 – 39.5	24	34.5	828
39.5 – 49.5	10	44.5	445
49.5 – 59.5	5	54.5	272.5
59.5 – 69.5	3	64.5	193.5
Total	63	—	2163.5

Mean For Grouped Data = $\frac{\Sigma fx}{\Sigma f} = \frac{2163.5}{63} = 34.34$

Another example:

mode = 3median - 2mean

32.5 = 3(?) - 2(34.341)

32.5 = 3(?) - 68.682

median = 33.72

Weaker responses revealed that the candidates could not use the given histogram to find the median. Instead, they tried to use formula of finding median from grouped data. Most of the candidates were able to fill the table and find the mean but again, for part iii, candidates did not know what empirical formula is and did a lot of guess work.

Example:

i. Estimate the mode of this data graphically. (1 Mark)

Mode lie between 29.5 – 39.5 So the value of Mode is (34)

ii. The frequency distribution of this histogram is shown in the given table. Fill in the table appropriately to find the mean of the data. (2 Marks)

Class Boundary	Number of Students	Class Mark	$\frac{f_i x_i}{\sum f_i}$
9.5 – 19.5	9	14.5	$\frac{9 \times 14.5}{63} = 2.071$
19.5 – 29.5	12	24.5	$\frac{12 \times 24.5}{63} = 4.666$
29.5 – 39.5	24	34.5	$\frac{24 \times 34.5}{63} = 13.142$
39.5 – 49.5	10	44.5	$\frac{10 \times 44.5}{63} = 7.063$
49.5 59.5	5	54.5	$\frac{5 \times 54.5}{63} = 4.325$
59.5 69.5	3	64.5	$\frac{3 \times 64.5}{63} = 3.071$
Total	63	—	34.338

Another example:

mode = 3median – 2mean mode = 3median – 2mean

2mean = 3median – mode 3median = 3mean – mode

2(34) 68.6824 – 34 3median = 2(34.3412) – 34

164.0472 3median = 34.6824

Question 4b:

The ages (in years) of a sample of 31 students were recorded. The results are shown in the following table.

Age (years)	Number of Students	Class Boundary
6 – 9	7	
10 – 13	8	
14 – 17	6	
18 – 21	10	
Total	31	—

Find the mode of this data.

Better responses reported that the candidates were able to identify modal group of the given data. The table was filled with class boundaries; correct formula of mode for grouped data was used; the values were substituted and the mode was found.

Example:

Age (years)	Number of Students	Class Boundary
6 – 9	7	5·5 - 9·5
10 – 13	8	9·5 – 13·5
14 – 17	6	13·5 - 17·5
18 – 21	10	17·5 – 21·5
Total	31	—

Find the mode of this data.

$$\hat{X} = L + \frac{(F_m - F_1)}{2F_m - F_1 - F_2} \times h$$

$$\hat{X} = 17.5 + \frac{(10 - 6)}{2(10) - 6 - 0} \times 4$$

$$\hat{X} = 17.5 + 0.285 \times 4$$

$\hat{X} = 18.64$ Answer

Weaker responses showed that the few candidates who wrote the formula correctly were unable to substitute the correct values of the symbols and failed to find the mode of the data. There were many who wrote incorrect formula. Some candidates found median instead of mode. The hierarchy of arithmetical operations also seemed to be a problem in the weaker responses.

Example:

Age (years)	Number of Students	Class Boundaries
6 – 9	7	5.5 – 9.5
10 – 13	8	9.5 – 13.5
14 – 17	6	13.5 – 17.5
18 – 21	10	17.5 – 21.5
Total	31	—

Find the mode of this data.

$$\text{Mode} = L + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$

$$\text{Mode class} = \frac{\Sigma f}{2} = \frac{31}{2} = 15.5$$

$$= 13.5 + \frac{6 - 8}{2(6) - 8 - 10} \times 4$$

$$\text{Mode} = 2.08$$

Question 5:

This was not a well attempted question.

Question 5:

A teacher designed a class project which was to be done by the entire class in groups of 5. Students of Group *A* managed to obtain full marks in the class project. After a certain time, the teacher sets up a 3-hour class test for the entire class and announced that the marks they obtain in this test will be added to the marks they had previously obtained in their class project.

The test marks obtained by the students of group *A* are 75, 88, 96, 52, 61.

i. Find the standard deviation of the test marks of group *A*.

ii. Each student of group *A* obtained 10 marks for the class project which will be added to their test marks. Would the standard deviation of the total marks after adding class project marks of group *A* be different from the standard deviation of the original test marks of this group? Support your answer with suitable property of measures of dispersion.

Better responses indicated that the candidates were able to understand the scenario and apply their skills in finding the standard deviation as per the given data. They correctly found the values of $\sum x$ and $\sum x^2$ substituted these values in the formula $\sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$ to find the required standard variance of the data. In part ii, candidates used the property of measures of dispersion $S.D(x+a) = S.D(x)$ to find the answer.

Example:

x	x^2
75	5625
88	7744
96	9216
52	2704
61	3721
372	29010

$$= \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$

$$= \frac{29010}{5} - \left(\frac{372}{5}\right)^2$$

$$= 5802 - (74.4)^2$$

$$= 5802 - 5535.36$$

$$\sigma^2 = 266.64$$

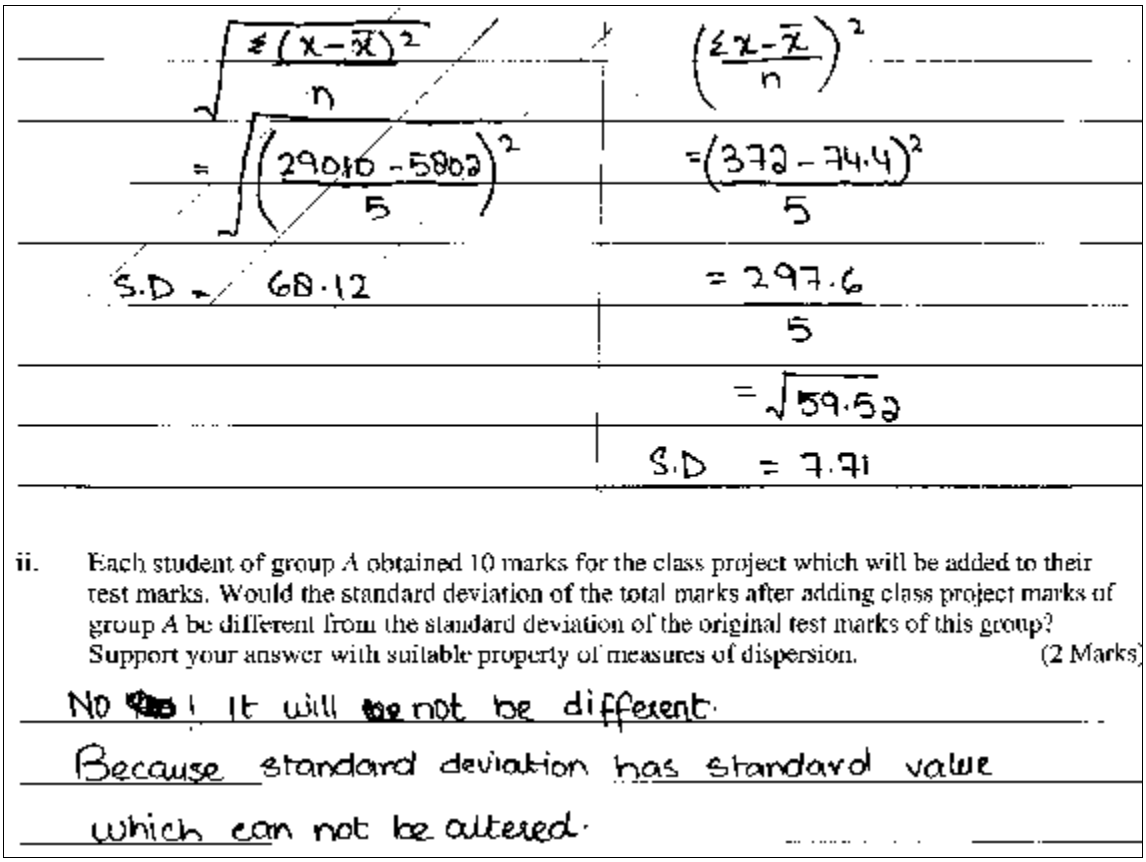
$$\sigma = 16.32$$

ii. Each student of group *A* obtained 10 marks for the class project which will be added to their test marks. Would the standard deviation of the total marks after adding class project marks of group *A* be different from the standard deviation of the original test marks of this group? Support your answer with suitable property of measures of dispersion. (2 Marks)

No, Standard deviation will not be different because by adding 10 marks each the Standard deviation become the same / constant = 16.32

Weaker responses indicated that the most of the candidates wrote the correct formula of the standard deviation but made mistakes in calculation of the values of $\sum x$ and $\sum x^2$. Moreover, they were unable to state the correct property to be used due to which they lost marks in the last part.

Example:



Question 6:

This question offered a choice between part a and part b. Majority of the candidates attempted part b. This was generally a well attempted question.

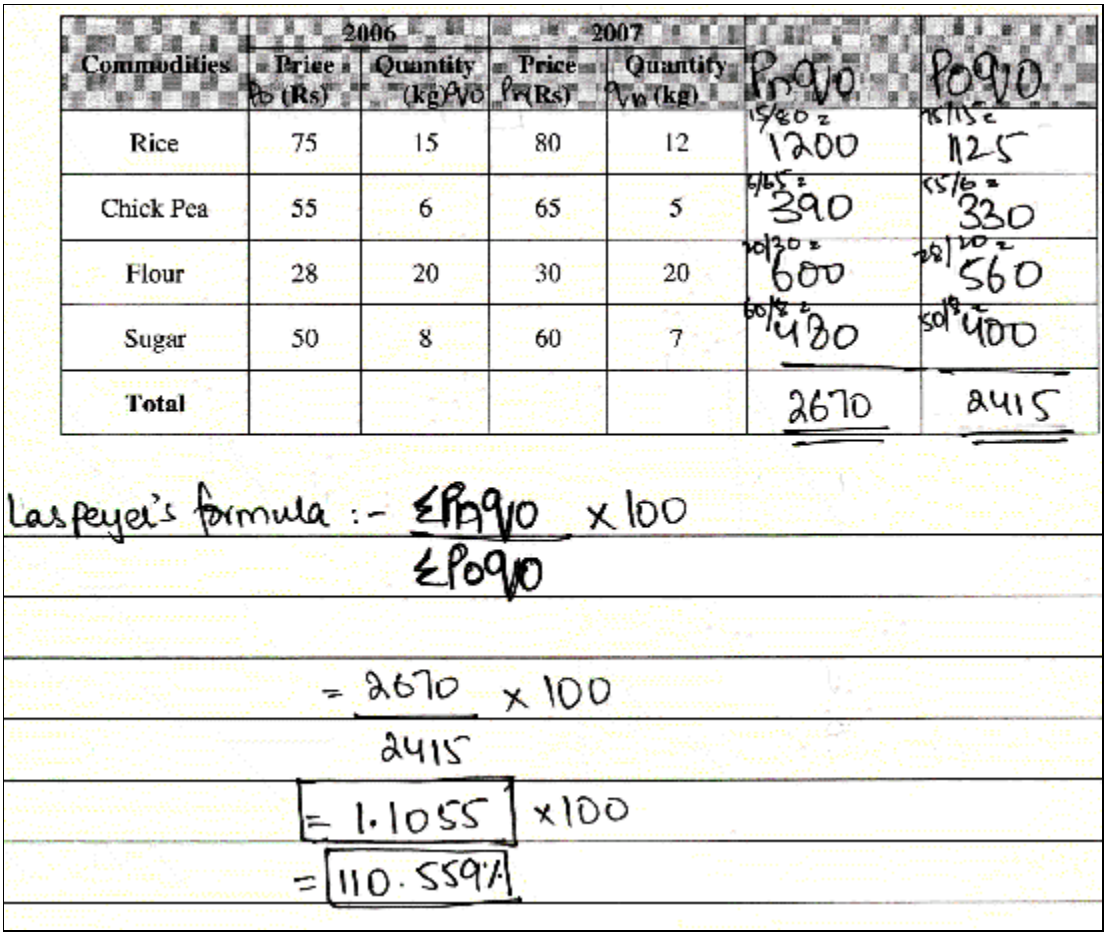
Question 6a:

In the given table, the prices (in Rupees per kilogram) and quantities (in kilogram) of some commodities in the years 2006 and 2007 are given. Find Laspeyer's index number, taking year 2006 as the base period.

Commodities	2006		2007			
	Price (Rs)	Quantity (kg)	Price (Rs)	Quantity (kg)		
Rice	75	15	80	12		
Chick Pea	55	6	65	5		
Flour	28	20	30	20		
Sugar	50	8	60	7		
Total						

Better responses indicated that the candidates had correct knowledge of Laspeyer's index number. From the given table, the values of $p_n q_o$ and $p_o q_o$ were found for each column. The total was substituted in the formula to find the correct answer.

Example:



Commodities	2006 Price p_o (Rs)	2006 Quantity (kg) q_o	2007 Price p_n (Rs)	2007 Quantity q_n (kg)	$p_n q_o$	$p_o q_o$
Rice	75	15	80	12	1200	1125
Chick Pea	55	6	65	5	390	330
Flour	28	20	30	20	600	560
Sugar	50	8	60	7	480	400
Total					2670	2415

Laspeyer's formula :- $\frac{\Sigma p_n q_o}{\Sigma p_o q_o} \times 100$

$= \frac{2670}{2415} \times 100$

$= 1.1055 \times 100$

$= 110.559\%$

Weaker responses displayed that most of the candidates were able to label the four columns as p_o, q_o, p_1 and q_1, but they could not figure out what was required in the last two columns. Use of incorrect formula was a frequent error too.

Example:

Commodities	2006 Price (Rs)	2006 Quantity (kg)	2007 Price (Rs)	2007 Quantity (kg)	p_1q_o	p_oq_o
Rice	75	15	80	12	900	1125
Chick Pea	55	6	65	5	275	330
Flour	28	20	30	20	560	560
Sugar	50	8	60	7	350	400
Total					2085	2415

Laspeyer's $= \frac{\Sigma p_o q_n}{\Sigma p_o q_o} \times 100$

$= \frac{2085}{2415} \times 100$

$= 86.335$

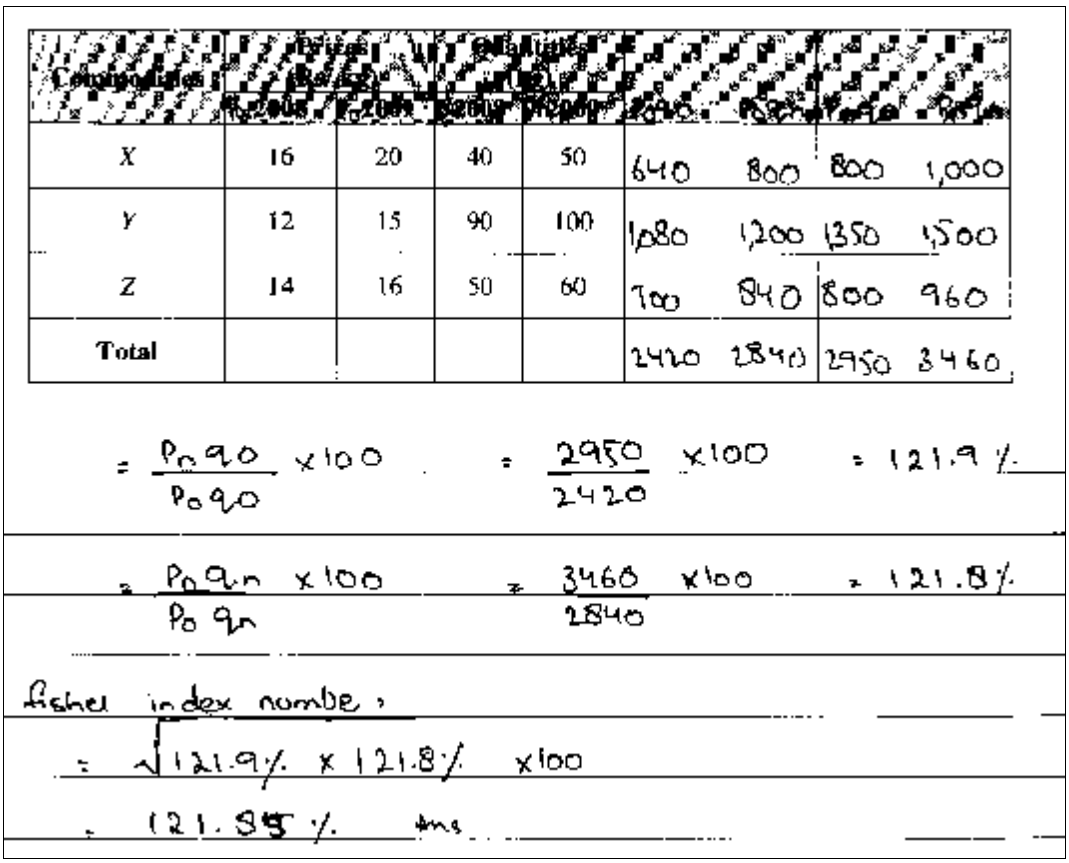
Question 6b:

In the given table, the prices (in Rupees per kilogram) and quantities (in kilogram) of three commodities *X*, *Y* and *Z* in the years 2008 and 2009 are given. Find Fisher's index number for the following data, taking year 2008 as the base period.

Commodities	**Prices (Rs/ kg)**		**Quantities (kg)**			
	2008	**2009**	**2008**	**2009**		
X	16	20	40	50		
Y	12	15	90	100		
Z	14	16	50	60		
Total						

Better responses indicated that the candidates correctly applied the method of finding Fisher's index number. The table was filled with $p_o q_o$, $p_1 q_o$, $p_0 q_1$ and $p_1 q_1$. The Paashes' formula was used to find the answer.

Example:



Commodities	P_0	P_1	q_0	q_1	P_0q_0	P_1q_0	P_0q_1	P_1q_1
X	16	20	40	50	640	800	800	1,000
Y	12	15	90	100	1080	1,200	1350	1,500
Z	14	16	50	60	700	840	800	960
Total					2420	2840	2950	3460

$= \frac{P_0 q_0}{P_0 q_0} \times 100 = \frac{2950}{2420} \times 100 = 121.9\%$

$= \frac{P_0 q_n}{P_0 q_n} \times 100 = \frac{3460}{2840} \times 100 = 121.8\%$

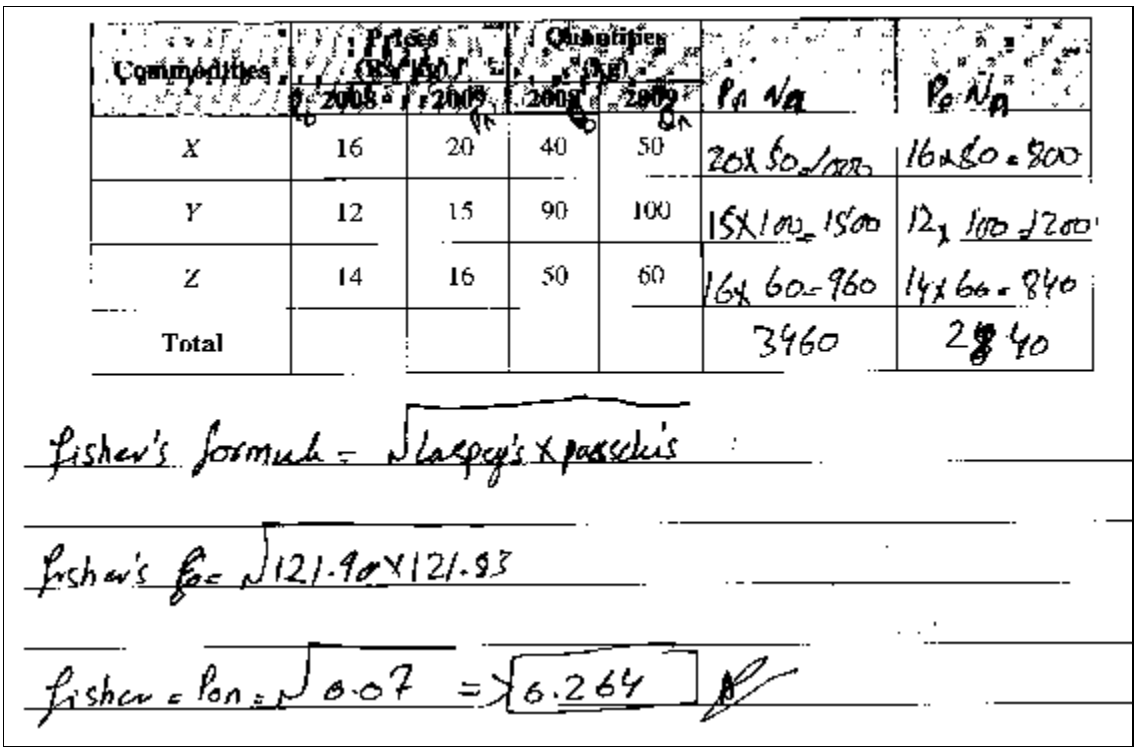
Fisher index number:

$= \sqrt{121.9\% \times 121.8\%} \times 100$

$= 121.85\%$ Ans

Weaker responses reflected that candidates lost marks due to the use of incorrect formula. Instead of Fisher's index method, candidates used Laspeyre's, Paasche's , and Marshall-Edgeworth's method. All such responses were not given credit.

Example:



Commodities	Prices (Rs/kg)		Quantities (kg)		$P_0 q_0$	$P_1 q_0$
	2008	2009	2008	2009		
X	16	20	40	50	20×50=1000	16×50=800
Y	12	15	90	100	15×100=1500	12×100=1200
Z	14	16	50	60	16×60=960	14×66=840
Total					3460	2840

Fisher's formula = $\sqrt{\text{Laspey's} \times \text{paasche's}}$

Fisher's $P_{01} = \sqrt{121.90 \times 121.93}$

Fisher = $P_{01} = \sqrt{0.07} = 6.264$

Question 7:

This question offered a choice between part a and part b. A majority of the candidates attempted part b. Candidates performed average in this question.

Question 7a:

Basit and Adnan both have a fair die with faces numbered 1, 2, 3, 4, 5 and 6. They throw their dice simultaneously.

(**Note:** Singular: die, Plural: dice)

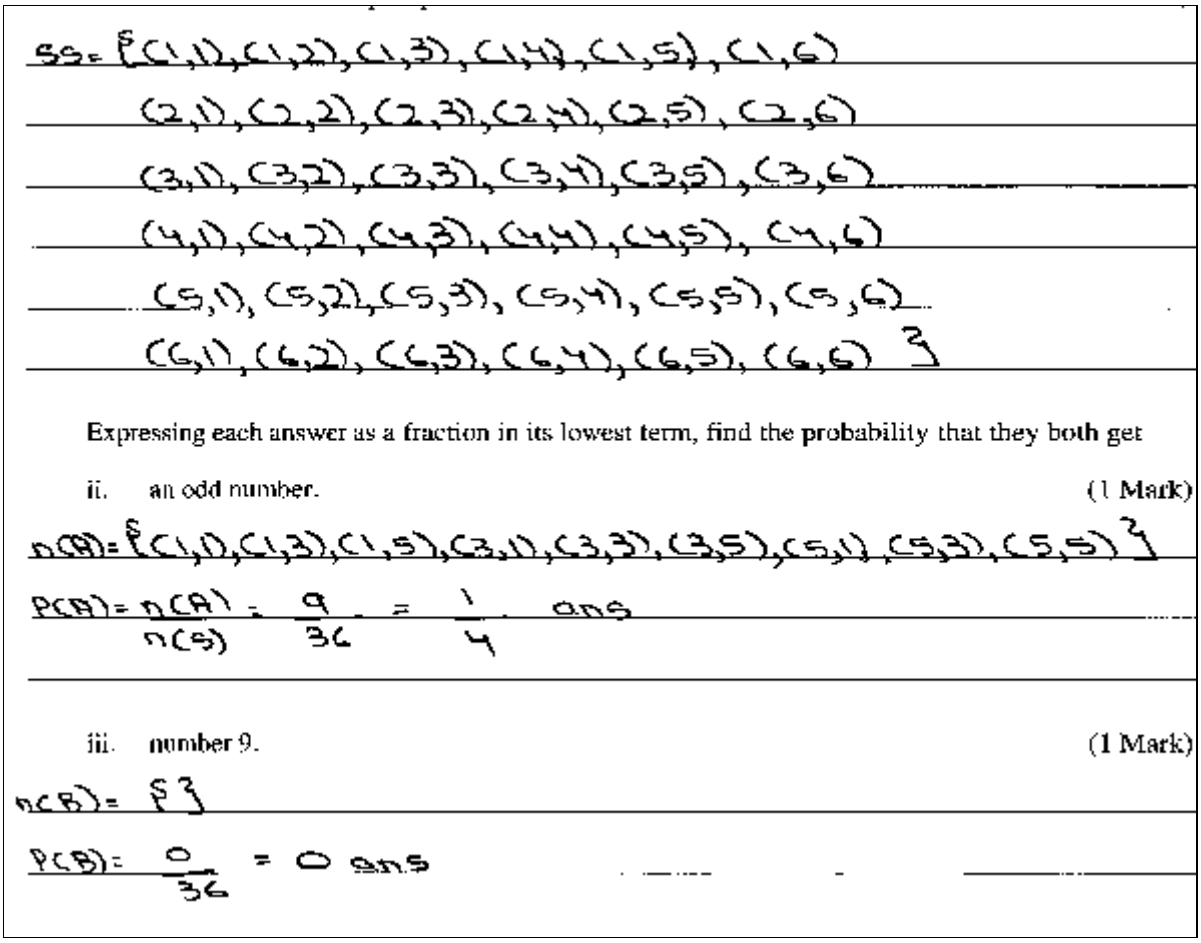
i. Write down the sample space.

   Expressing each answer as a fraction in its lowest term, find the probability that they both get

ii. an odd number.
iii. number 9.

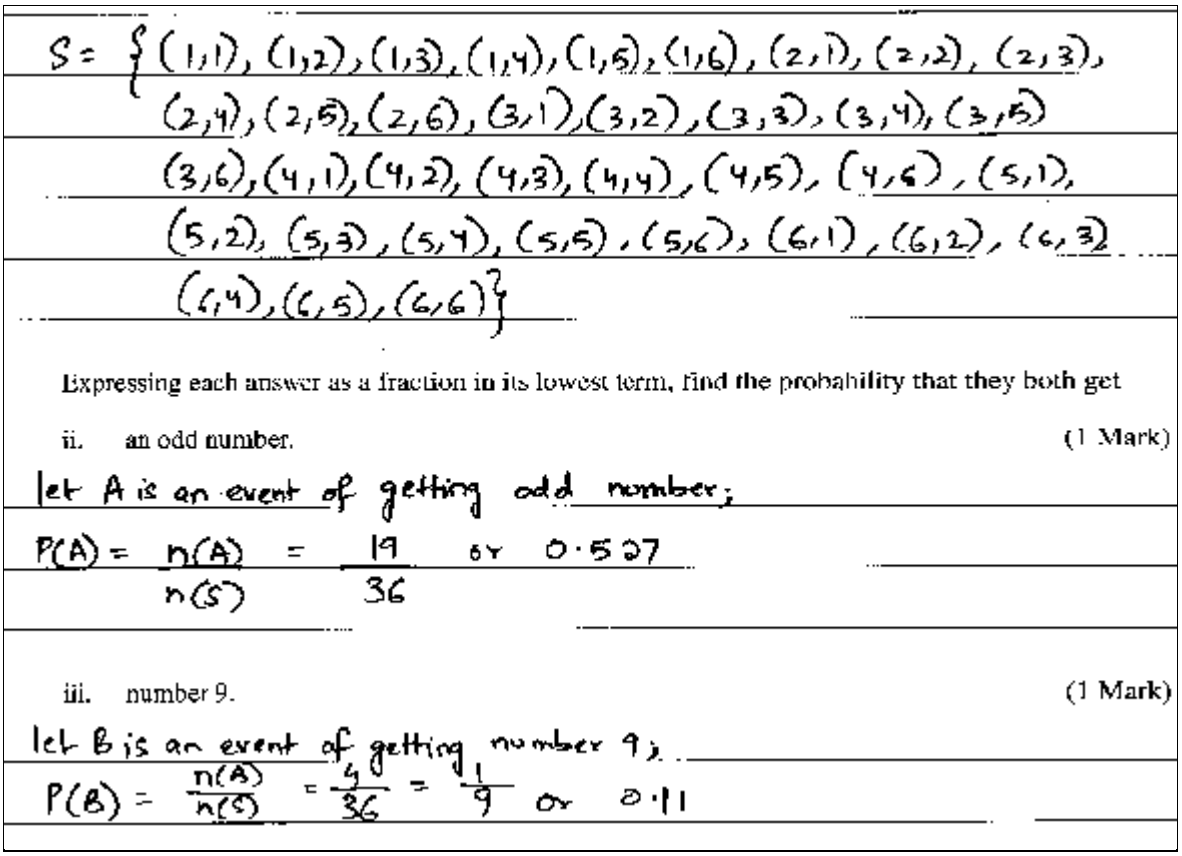
Better responses indicated that the candidates were able to understand the given experiment and were able to form the sample space. Some candidates made a tree diagram to help them find the sample space. For part ii, candidates used the sample space/ tree diagram to find the probability of getting an odd number. A smaller set was chosen that depicted this particular condition and the probability was found. Candidates easily attempted part iii, which shows their understanding that probability ranges from 0 – 1.

Example:



Weaker responses displayed that many candidates failed to infer that if Basit and Adnan take turns to throw die, it means that both these experiments are happening one after another and the sample space would be a combination of the outcomes of both. Hence they listed the outcomes of single throw and lost marks. In part ii and iii, some candidates were able to list sample points of the events but failed to find the probabilities of the events.

Example:



$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$

Expressing each answer as a fraction in its lowest term, find the probability that they both get

ii. an odd number. (1 Mark)

let A is an event of getting odd number;

$P(A) = \frac{n(A)}{n(S)} = \frac{19}{36}$ or 0·527

iii. number 9. (1 Mark)

let B is an event of getting number 9;

$P(B) = \frac{n(A)}{n(S)} = \frac{4}{36} = \frac{1}{9}$ or 0·11

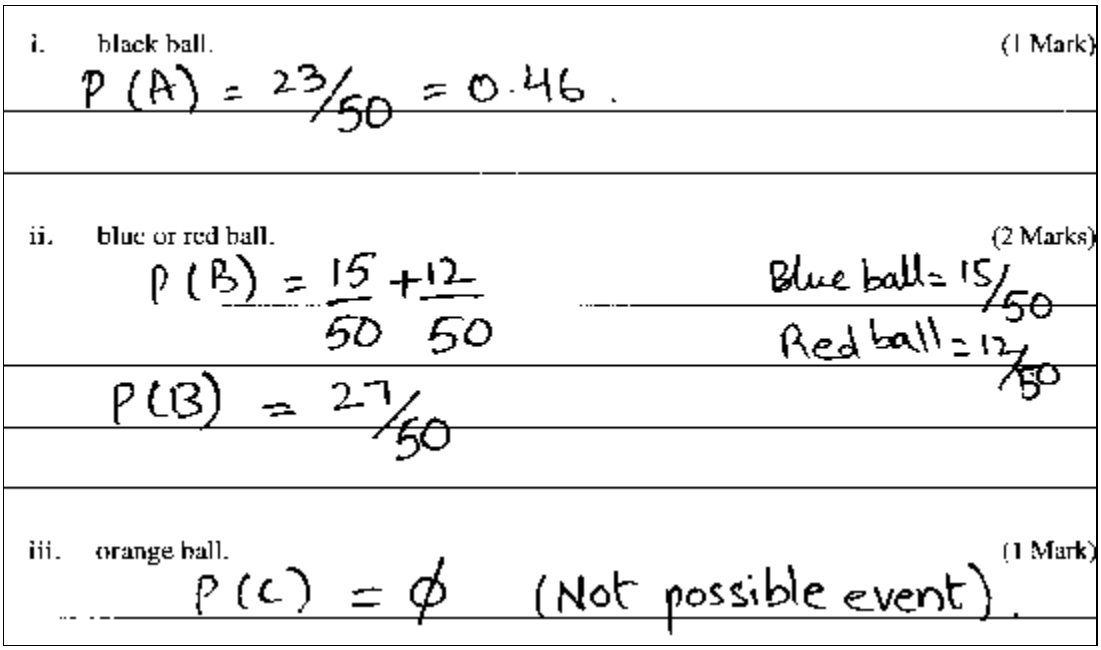
Question 7b:

A box contains 15 blue balls, 12 red balls and 23 black balls. If Sara draws a ball at random from this box, find the probability it is a/ an

i. black ball.
ii. blue or red ball.
iii. orange ball.

Better responses exhibited that for part i, candidates found the total number of balls and found the probability of finding the black ball. In part ii, candidates were able to understand the 'or' situation, found probability of blue and red ball separately and added these two. In part iii, candidates exhibited their knowledge that the probability of a possibility that does not exist in sample space is zero.

Example:



Weaker responses reflected weak concepts of probability. Most of the candidates were able to do part i but failed to do part ii. In part ii, the most frequent incorrect response was $\frac{15}{50}\times\frac{12}{50}$ which shows that candidates failed to determine the implication of the use of 'or'.

Example:

